

Per-Query Environmental Accounting for Local-First AI: A Verifiable Methodology

Matthew J. Ernewein

Hyacinth AI, Inc.

`matthew@hyacinth.ai`

Mark Somol

Hyacinth AI, Inc.

`mark@hyacinth.ai`

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Abstract

This paper presents two contributions: (1) an accounting methodology for computing per-query environmental impact in local-first artificial intelligence applications, and (2) a disclosure paradigm — transaction-level environmental receipts — that we argue addresses a structural gap in current sustainability reporting for AI systems. The methodology produces a *receipt* comprising the energy and water consumed and the carbon-equivalent emissions produced by a single query, alongside a defensible comparison to the equivalent query routed through a cloud-hosted large language model. The framework distinguishes user-attributable costs (compute on the user’s device, transport energy on the user’s network adapter) from upstream costs (data center cooling, indirect grid generation, third-party data center compute) and applies industry-published coefficients with documented ranges. We describe a reference implementation in Mnemne, a local-first knowledge application, where the receipt is surfaced both per query and as a cumulative running total, and we discuss how the framework can be parameterized for other local-first systems. We argue that user-verifiable per-query disclosure — analogous to itemized receipts in financial transactions or per-meal nutrition labeling — offers methodological transparency at the level at which user actions are actually taken.

Keywords: sustainable AI; local-first computing; environmental accounting; large language models; on-device inference.

1. Introduction

Artificial intelligence has become a meaningful contributor to global electricity consumption, with frontier-class large language model (LLM) inference dominating the per-query energy footprint of modern AI products. Concurrent reporting on the water footprint of cloud data centers has surfaced consumption figures of hundreds of millions of liters annually attributable to a single hyperscaler region. As these costs become tangible, non-expert users increasingly want to understand the environmental impact of the AI services they use.

The dominant industry response has been *aggregate* reporting: companies publish annual sustainability disclosures with composite metrics — power-usage effectiveness (PUE), water-usage effectiveness (WUE), total grid carbon intensity — that are accurate at the level of an entire data center portfolio but not directly legible at the per-query level the user actually generates. A user posing a single question to an AI service has no first-party method to estimate *that* query’s marginal contribution to consumption.

This paper contributes both an accounting methodology and a disclosure paradigm. The accounting methodology computes per-query energy, water, and carbon-equivalent emissions from observed inputs (tokens generated, bytes transferred) using industry-published coefficients with documented ranges; the disclosure paradigm renders those quantities to the user as an itemized receipt accompanied by a comparison to an equivalent cloud query, with the methodology sufficiently transparent that a technically sophisticated user can recompute every line. The accounting is conventional in form; the disclosure framing — transaction-level receipts as the unit of sustainability reporting, rather than annual aggregates — is the contribution we believe is novel.

We frame the methodology around three goals:

1. **Per-query granularity.** Each query produces a self-contained receipt computed from observed inputs, not extrapolated from aggregates.
2. **User-attributable accounting.** We distinguish costs the user directly incurs (compute on their hardware) from costs imposed on the wider system (upstream data center compute, ISP backbone) and apply different treatment to each.
3. **Methodological transparency.** Coefficients are published, sourced, and ranged. The receipt’s components are exposed in a methodology page so a sophisticated user can audit the math.

The intended outcome is a per-query *savings* number — the difference between Mnemne’s local cost and an equivalent cloud query — which we display alongside the original receipt. Selecting the right cloud baseline is essential to the honesty of this comparison: matching a Mnemne agentic-pipeline query against a single-shot cloud chat query would understate the cloud baseline by a factor of five to eight and produce a misleading “savings” figure (Section 3.3.2). Section 3 documents the cloud-baseline derivation explicitly so a reviewer can audit the matching choice as well as the coefficients.

Scope. The reference implementation described here is specific to Mnemne (Phi-4-mini chat model, nomic-embed-text v1.5 embedding model, Apple Silicon GPUs, llama.cpp inference engine). Section 7 discusses how the framework generalizes to other local-first systems with different hardware and model choices.

The remainder of this paper proceeds as follows. Section 2 surveys prior work on AI and ICT sustainability accounting. Section 3 documents the coefficients and computation framework, including explicit derivations for cloud baselines and a treatment of heating/cooling-climate effects on local PUE. Section 4 walks through worked examples for the two query types Mnemne supports (library search over local documents, agentic web search). Section 5 discusses limitations and confidence intervals. Section 6 develops the “sustainability-as-receipts” disclosure paradigm in more detail, including treatment of the objection that approximate per-query estimates do not warrant transaction-level disclosure. Section 7 sketches the parameterization required to apply this framework to other systems. Section 8 concludes.

2. Related Work

The dominant academic baseline for LLM energy accounting is the training-time analysis of Patterson et al. [1], which decomposed the greenhouse-gas footprint of large neural network training into hardware, location, and algorithm components and surfaced PUE-adjusted total emissions for several published models. Strubell et al. [2] introduced the practice of expressing energy in comparable, real-world units (US household-equivalent days, transatlantic flight equivalents), now standard.

Inference-time analyses have been comparatively rare until recently. Luccioni et al. [3] documented per-query energy for several open-weight models across modalities, demonstrating that a single inference of a multi-billion-parameter LLM consumed measurable but variable energy. The Hugging Face AI Energy Score program [4] extended this to standardized per-query benchmarks across deployed model endpoints, providing the first publicly cited per-query baseline for “a typical cloud LLM query” on dense open-weight architectures. For closed frontier models, which use mixture-of-experts designs and do not publish per-query disclosures, Epoch AI [5] provides an independent per-query estimate (GPT-4-class, ≈ 0.30 Wh); Oviedo et al. [18] independently derive a median of 0.34 Wh (preprint) and 0.31 Wh (*Joule*) for frontier-scale ($> 200\text{B}$ -parameter) models on H100 using a bottom-up production-batched methodology; and OpenAI’s CEO has publicly stated ≈ 0.34 Wh per query [17]. We anchor the cloud baseline on the convergence of these three estimates.

The broader information-and-communications-technology (ICT) energy footprint literature is more mature. Malmodin and Lundén [13] published end-to-end accounting for the global ICT sector including operator networks, end-user devices, and data centers, arriving at substantially more conservative network energy coefficients than earlier work. Hertwich and Wood [14] provide the broader scope-3 industrial-emissions framing within which electricity-driven workloads sit. The narrower question of when to use *marginal* versus *average* grid carbon intensity — which matters

when an AI workload is *additional* load on a grid (marginal intensity) versus *replacing* existing load (average intensity) — is treated in the electricity-emissions methodology literature [15]. Local-first AI is best modeled as additional load in both cases, so the average/marginal distinction is less pivotal than for, say, EV charging time-shifting; we use grid-average intensity throughout and document the assumption.

Water-footprint analysis of AI compute is newer. Sustainability disclosures of major hyperscaler operators (e.g., Google [7]) publish water-usage-effectiveness (WUE) figures distinguishing direct cooling-tower consumption from indirect generation-grid consumption. On-site versus supply-chain attribution remains an active area of methodological debate.

To our knowledge, no published framework has combined these inputs into a *per-user*, *per-query*, *on-device receipt* with a comparison to cloud baselines. The framework documented here is intended to fill that gap for local-first applications.

3. Methodology

3.1 Components

A query receipt is composed of four input streams, each measured in its native unit and converted to a common comparison basis (energy in watt-hours, water in milliliters, carbon-dioxide-equivalent in grams):

1. **Local compute energy.** Tokens generated and prompt tokens consumed by the bundled chat and embedding models, multiplied by measured per-token energy coefficients.
2. **Local network energy.** Bytes transferred over the user’s network adapter, multiplied by a per-megabyte energy coefficient representing the active power draw of the network interface during transfer.
3. **Cloud baseline.** A per-query midpoint estimate selected by query type (single-shot LLM versus agentic-pipeline LLM), with explicit derivation in Section 3.3.
4. **Conversion factors.** Power-usage-effectiveness (PUE) multipliers to translate IT-load energy into wall-plug equivalent; water-usage coefficients to translate kWh into water consumption via grid-generation and direct-cooling pathways; grid carbon intensity to translate kWh into grams of CO₂-equivalent.

Each component is published with a midpoint and an explicit confidence range. Section 5 documents these ranges and how they propagate.

3.2 Local Compute Energy

For the bundled chat model (Phi-4-mini, $\approx$ 3.8B parameters in 4-bit quantization), running on Apple Silicon GPUs via the `llama.cpp` inference engine, sustained GPU power draw under inference is approximately 12 watts (mid-tier M2/M3 systems, measured via `powermetrics`). Generation throughput averages 50 tokens per second. Per-token energy in watt-hours is therefore:

$$E_{\text{gen}} = \frac{P_{\text{GPU}}}{R_{\text{gen}} \cdot 3600} = \frac{12}{50 \cdot 3600} \approx 6.7 \times 10^{-5} \text{ Wh/tok}$$

For prompt evaluation (the parallel attention pass that consumes input tokens before generation begins), throughput is approximately 1000 tokens per second at the same GPU load:

$$E_{\text{prompt}} = \frac{12}{1000 \cdot 3600} \approx 3.3 \times 10^{-6} \text{ Wh/tok}$$

The bundled embedding model (`nomic-embed-text v1.5` in 8-bit quantization, $\approx 137\text{M}$ parameters) processes a chunk (typically 256–512 tokens) in approximately 3 ms, yielding:

$$E_{\text{embed}} = \frac{12 \cdot 0.003}{3600} \approx 1.0 \times 10^{-5} \text{ Wh/chunk}$$

Local PUE and the heating/cooling interaction. Cloud data center PUE varies considerably by operator and vintage: Google’s fleet-wide trailing-twelve-month PUE was 1.10 [7], while the broader industry average across all surveyed data centers was 1.58 in 2023 [12]. AI workloads concentrate in hyperscaler facilities at the lower end of this range, but newer high-density AI buildouts and non-hyperscaler colos pull the average upward. We adopt a midpoint of ≈ 1.4 , with the full 1.10–1.58 range carried through the sensitivity analysis in Section 5.

Local PUE requires different treatment. (PUE in its standard usage is a data center metric; we extend the term here as an analogous on-device coefficient, since the same overhead-on-IT-load logic applies to a personal device dissipating compute heat into the surrounding room.) The user’s device dissipates compute heat into the surrounding room air; whether that constitutes overhead depends on whether the room is being heated or cooled at the time:

- In **heating-dominant climates and seasons** (e.g., a Massachusetts apartment from October through April), waste heat from local compute displaces approximately 95% of an equivalent unit of heating-system delivered heat. Effective $\text{PUE}_{\text{local}}$ is therefore *less than 1.0* relative to a counterfactual where the heating system would have done that work. The exact value depends on the heating system’s coefficient of performance: for resistive electric heat (COP 1), 1 Wh of compute waste heat displaces 1 Wh of heating electricity, giving $\text{PUE}_{\text{local}} \approx 1 - 0.95 = 0.05$ (and 0 in the limiting case of full thermal capture); for a heat pump at COP 3, 1 Wh of waste heat displaces only ≈ 0.33 Wh of heat-pump electricity, giving $\text{PUE}_{\text{local}} \approx 1 - 0.95/3 \approx 0.68$. We do not currently credit the user with this rebate.
- In **cooling-dominant climates and seasons**, waste heat increases the air-conditioning load. With a typical AC coefficient-of-performance of 3, every joule of compute heat costs an additional ≈ 0.33 J of AC compressor work to remove. Effective $\text{PUE}_{\text{local}}$ is therefore approximately 1.33.
- In **temperate climates and shoulder seasons**, neither effect applies; $\text{PUE}_{\text{local}} \approx 1.0$.

We default to $\text{PUE}_{\text{local}} = 1.0$ for simplicity, which is the temperate-climate midpoint. A future extension would use the user’s IP-derived climate zone and current-month seasonality to apply a more accurate adjustment. The bias from this simplification is near-zero on annual aggregate for users in mid-latitude climates and moderately conservative for users in heating-dominant climates.

Total local compute energy for a query is the sum across all model calls of the per-token and per-chunk products, multiplied by the applicable $\text{PUE}_{\text{local}}$.

3.3 Cloud Baseline (Explicit Derivation)

We use two distinct cloud baselines depending on the query type. Both are derived from publicly available sources, with the derivation spelled out below to satisfy the verifiability standard the rest of this paper aims for.

3.3.1 Single Cloud LLM Query: 0.34 Wh

Three independent public sources now converge on the per-query energy of a frontier cloud chat call. Epoch AI [5] estimates a single GPT-4-class query at ≈ 0.30 Wh, reflecting the mixture-of-experts (MoE) architecture used by closed frontier models — an architecture in which only a small subset of the model’s total parameters is activated for any given token, so per-query energy scales with active rather than total parameter count. OpenAI’s CEO publicly stated ≈ 0.34 Wh per ChatGPT query in June 2025 [17]. Independently, Oviedo et al. [18] derive a median of 0.34 Wh in the preprint and 0.31 Wh in the published *Joule* version (April 2026) for frontier-scale ($> 200\text{B}$ -parameter) models on H100s using a bottom-up production-batched methodology, with an interquartile range of 0.18–0.67 Wh; the slight downward revision sits between Epoch’s and OpenAI’s figures and does not change the qualitative picture. Hugging Face’s AI Energy Score [4] reports per-query benchmarks for *dense* open-weight models on standardized hardware (Text Generation task, model class $> 66\text{B}$ parameters); Llama-3-70B-class results land at ≈ 1.7 Wh per query — substantially higher than MoE frontier models because the dense architecture activates every parameter on every token, and because the benchmark is deliberately measured at batch size 1 for fair cross-model comparison (see Section 3.3.3).

Most cloud chat traffic for the use cases Mnemne targets (general research and synthesis) routes to closed MoE frontier models (GPT-4o-class, Claude 3.5-class, Gemini 1.5 Pro-class) rather than dense 70B-class hosted inference. We therefore adopt **0.34 Wh** as the single-query midpoint — the convergent figure across Epoch AI’s 0.30 Wh estimate, OpenAI’s 0.34 Wh disclosure, and Oviedo et al.’s independently derived median of 0.34 Wh (preprint) and 0.31 Wh (*Joule*) — with a documented range of 0.20–0.50 Wh. Note that if the cloud counterfactual were a dense 70B-class endpoint instead, savings against Mnemne would be roughly $6\times$ larger; the choice we have made is conservative.

3.3.2 Cloud Agentic Search: 2.0 Wh

A cloud agentic search pipeline (Perplexity Pro, ChatGPT with web browsing, Gemini Deep Research) typically performs a multi-step plan-execute-synthesize workflow [21, 22]. The decomposition below is our estimate based on publicly documented architectures:

- 1 query-rewriting / planning LLM call
- 1 search call (modest energy; metadata only)
- 3–6 page-content distillation LLM calls (one per fetched URL)
- 1 synthesis LLM call

This corresponds to **5–8 frontier-class LLM calls** plus retrieval. Applying the 0.34 Wh per-call midpoint yields:

$$E_{\text{agentic, low}} = 5 \cdot 0.34 = 1.7 \text{ Wh}, \quad E_{\text{agentic, high}} = 8 \cdot 0.34 = 2.72 \text{ Wh}$$

We adopt **2.0 Wh** as a conservative round midpoint, with a documented range of 1.7–2.7 Wh. The relative-comparison framing in Section 5 holds across this range.

3.3.3 Note on Batching Efficiency

A reviewer familiar with GPU inference economics may observe that cloud data centers batch concurrent queries to amortize fixed kernel and memory-loading costs across users, which a single-user local Mac cannot. This effect is *already incorporated* into the 0.34 Wh midpoint we use: the anchoring sources for the cloud baseline — Epoch AI’s GPT-4-class estimate [5] and Oviedo et al.’s bottom-up derivation [18] — model production-condition deployments that already perform batching at production volume. (The Hugging Face AI Energy Score figures [4], by contrast, are deliberately measured at batch size 1 on H100s for fair cross-model comparison and are *not* used as the cloud baseline; that benchmarking choice is part of why the dense 70B-class number lands at ≈ 1.7 Wh.) We are therefore not overstating the cloud baseline by ignoring batching efficiency — the published per-query figure implicitly reflects it. If anything, our cloud baseline is slightly favorable to cloud (higher than what an idealized batched endpoint would consume), which biases the comparison in favor of cloud and against Mnemne. Savings figures should be read as conservative on this axis.

Cloud baselines are PUE-adjusted by $\text{PUE}_{\text{cloud}} = 1.4$, a midpoint between hyperscaler best-in-class (Google fleet-wide 1.10 [7]) and broader industry average (1.58 across all surveyed data centers in 2023 [12]), representing the additional wall-plug energy consumed by data center cooling and infrastructure beyond the IT load. The 1.10–1.58 range is carried through Section 5.

Selecting the *right* cloud baseline is essential to honest comparison. Comparing Mnemne’s eight-LLM-call agentic web pipeline to a single cloud chat query would understate the cloud baseline by a factor of 5–8 and present a misleading “savings” figure. The correct comparison is to a cloud product performing equivalent work, which is the framing we use throughout.

3.4 Water Consumption

Water consumption per kWh of electricity is the sum of *direct* (cooling tower evaporation) and *indirect* (grid-generation cooling at thermal plants) pathways:

$$W_{\text{cloud}} = W_{\text{cooling}} + W_{\text{generation}}$$

For cloud data centers, $W_{\text{cooling}} \approx 1.0$ L/kWh, the figure originally derived in the 2016 Lawrence Berkeley National Laboratory *United States Data Center Energy Usage Report* [6] and quoted widely downstream. Hyperscaler-published WUE is in many cases substantially lower (Microsoft’s FY24 global WUE is 0.30 L/kWh, with their Americas regional figure at 0.55 L/kWh [19]); the reconciliation with Altman’s figure later in this section returns to this point. We retain the LBNL coefficient because adopting the hyperscaler best-case here would inflate Mnemne’s apparent water savings, since cloud is the term being subtracted from. The grid-generation term $W_{\text{generation}}$ is the consumption-side US average, computed by weighting per-technology consumptive water-use factors from NREL [9] against the current US generation mix [8]; that derivation lands near 1.4 L/kWh. We note that this is at the low end of the published range — peer-reviewed estimates of US-grid water consumption span roughly 1.4–3.14 L/kWh depending on whether hydroelectric reservoir evaporation is included [10, 11], and the canonical AI-water paper by Li et al. adopts 3.14 L/kWh from Reig et al. as its US-average figure. We retain the lower derived value because using the higher 3.14 L/kWh figure would inflate cloud’s footprint and overstate Mnemne’s water savings; the choice is conservative against Mnemne.

The cloud-side midpoint is therefore the simple sum of the two components: $W_{\text{cloud}} = 1.0 + 1.4 = 2.4$ L/kWh. Higher totals circulate in the trade press — often reaching 3.0 L/kWh or more, weighted toward hot-region data centers with higher evaporative cooling loads and older facilities — but we do not adopt them because doing so would inflate cloud’s footprint relative to Mnemne and overstate the savings claim this paper defends.

For local compute, $W_{\text{cooling}} \approx 0$ (air cooling, no water in the loop) and $W_{\text{generation}} \approx 1.4$ L/kWh per the same derivation, yielding 1.4 L/kWh.

Per-query water is computed by multiplying the query’s energy in kWh by the appropriate water-intensity coefficient:

$$V_{\text{water,query}} = E_{\text{query,kWh}} \cdot W$$

Reconciliation with Altman’s ≈ 0.32 mL/query figure. A widely circulated number from OpenAI’s CEO [17] places per-query water consumption at 0.000085 gallons (≈ 0.32 mL) for an average ChatGPT call — substantially below the ≈ 1.15 mL per cloud query implied by our coefficients ($0.34 \text{ Wh} \cdot 1.4 \cdot 2.4 \text{ L/kWh}$). The two numbers are consistent under different scope choices: Altman’s figure plausibly reflects only *direct on-site cooling* at hyperscaler best-in-class

WUE (Microsoft FY24’s 0.30 L/kWh global figure; Microsoft Americas at 0.55 L/kWh [19]) and excludes the indirect generation-side water embedded in the electricity the data center consumes; our figure includes both. A reviewer using Altman’s narrower scope would shrink both sides of our comparison without changing its sign — local on-device cooling is also ≈ 0 in the Mnemne case (air-cooled, no water in the loop), so the 0-vs- ≈ 0.3 mL inequality still favors Mnemne, just by a smaller absolute margin.

3.5 Carbon Dioxide Emissions

Per-query CO₂-equivalent emissions are computed as the product of energy and grid carbon intensity:

$$M_{\text{CO}_2, \text{query}} = E_{\text{query, kWh}} \cdot I_{\text{CO}_2}$$

We use a US-grid-average value of $I_{\text{CO}_2} = 380$ g CO₂e/kWh [8] as the default. Following standard practice for additional loads on the grid [15, 14], we use grid-average rather than marginal intensity: local-first AI is *additional* load (it does not directly displace existing load) but the marginal versus average distinction is largely insensitive at the per-query scale and within the uncertainty bands of other coefficients in the methodology. Regional overrides via runtime geolocation-to-grid-mix lookup against electricityMaps or a similar service are planned. The default applies the same carbon intensity to both local and cloud sides. We explicitly do not claim this is conservative against Mnemne: cloud data centers are increasingly sited in lower-carbon regions while user devices are powered by their actual residential grid, so a genuinely conservative treatment would apply a lower intensity on the cloud side than on the local side. We choose the symmetric single-coefficient treatment for tractability and transparency, not for directional advantage, and flag the asymmetry here so a reviewer applying region-aware coefficients can recompute.

3.6 Network Energy

The methodology accounts only for *user-side* network transport energy: the active power draw of the user’s WiFi or wired interface during the seconds it spends transferring bytes for a given query. For a typical Apple Silicon Mac’s WiFi interface drawing ≈ 1 watt during active download, transferring 1 megabyte at typical residential broadband speeds (≈ 100 Mbps) takes approximately 80 ms, yielding:

$$E_{\text{network}} \approx \frac{1 \text{ W} \cdot 0.08 \text{ s}}{3600 \text{ s/h}} \approx 2.2 \times 10^{-5} \text{ Wh/MB}$$

The full-stack ICT energy attribution literature is significantly broader: Andrae [16] estimates 0.06–1.0 kWh/GB depending on attribution methodology, while Malmmodin and Lundén [13] arrive at substantially more conservative figures with more careful end-use attribution. The methodology exposes the raw bytes-transferred field on the network log so users or auditors can apply their

preferred network-energy model externally. The relative comparison between local and cloud queries is comparatively insensitive to the choice of network-energy coefficient: both cases traverse broadly similar public internet and backbone infrastructure, so first-order transport comparisons depend only weakly on the exact per-megabyte coefficient chosen. We acknowledge that cloud agentic pipelines may additionally incur internal east-west data center traffic, replication, and proprietary retrieval-layer transport that have no local counterpart; these are excluded from both sides for tractability and discussed under “Upstream cloud-side data center compute” in Section 3.7.

3.7 Excluded Costs

For completeness we enumerate costs not currently included in the receipt:

- **Local vector database operations.** Mnemne’s vector store (LanceDB) executes a small number of disk reads and a CPU-side nearest-neighbor scan per query. Measured energy is dominated by momentary CPU activity ($\approx 5\text{--}10$ W for < 100 ms per query), yielding $\approx 1.5 \times 10^{-4}$ Wh per query — approximately one part in 700 of a typical library query’s total. We exclude it from the receipt for tractability and to keep the receipt formula composable across vector-store implementations.
- **Embodied carbon of the user’s device.** Manufacturing carbon ($\approx 150\text{--}250$ kg CO₂e for a typical Mac) is not attributed to AI workloads, since the device is purchased for general personal computing; AI is incremental. Manufacturing carbon for cloud data center hardware is similarly excluded from the cloud baseline. We invite users with stricter accounting requirements to add embodied carbon proportionally. Note that this exclusion is not symmetric in Mnemne’s favor: a user’s Mac amortizes its embodied carbon across all general-purpose computing, of which AI is a small slice, whereas a purpose-built AI rack in a hyperscaler facility amortizes its embodied carbon only across AI workloads. A reviewer reintroducing embodied carbon proportionally would therefore add more to the cloud per-query figure than to the local one.
- **Backbone and ISP transport.** As noted in Section 3.6.
- **Upstream cloud-side data center compute for web search.** Mnemne’s “Ask the web” feature queries third-party search engines (Google, Bing, DuckDuckGo, etc.) which themselves run data center workloads to serve results. We do not attribute this cost because
 - (1) the same upstream cost is incurred by any search frontend including cloud agentic alternatives, so it cancels in the comparison, and (2) reasonable attribution methodologies disagree by more than an order of magnitude. There is, however, a small *asymmetry* worth acknowledging: as a metasearch, Mnemne’s SearXNG layer fans out each query to multiple upstream engines (Google, Bing, DuckDuckGo, Brave, Wikipedia, ...), whereas a cloud agentic product such as Perplexity may query only its own index or a single contracted search API. The upstream search-serving cost for a Mnemne web query is therefore moderately higher than for the cloud comparator on a per-query basis. The most widely cited per-search energy figure — Google’s own ≈ 0.3 Wh per query, disclosed by Urs Hölzle in 2009 [20] — implies that a five-engine fan-out adds on the order

of ≈ 1.5 Wh of upstream search-serving energy versus a single contracted index. Set against an LLM-inference term on the order of 2 Wh per agentic pipeline, this is the same order of magnitude, not negligible. We note three mitigating factors: (1) the 2009 figure predates fifteen years of data center and search-stack efficiency improvements and is likely an upper bound today; (2) cloud agentic products such as Perplexity also fan out, typically across multiple sources internally, so the asymmetry is smaller than the worst-case framing implies; and (3) the receipt-as-disclosure framing of this paper means a user concerned about upstream search-serving energy can recompute with their own preferred coefficient. We acknowledge the asymmetry as a real qualifier on the savings claim rather than a strict order-of-magnitude domination.

3.8 Receipt Composition

The final receipt for a query is computed as:

$$E_{\text{Mnemne}} = (\sum E_{\text{models}} + E_{\text{network}}) \cdot \text{PUE}_{\text{local}} \quad (1)$$

$$E_{\text{cloud}} = E_{\text{baseline}} \cdot \text{PUE}_{\text{cloud}} \quad (2)$$

$$E_{\text{saved}} = \max(0, E_{\text{cloud}} - E_{\text{Mnemne}}) \quad (3)$$

Water and carbon savings follow analogously. The receipt is rendered to the user as a per-query line containing:

- Energy saved in watt-hours, with a comparison string (“ ≈ 6 minutes of a 10 W bulb”)
- Water saved in milliliters, with a comparison string (“ ≈ 1 teaspoon”)
- CO₂-equivalent saved in grams, with a real-world equivalent

A separate aggregate dashboard accumulates savings across daily, monthly, and lifetime windows.

4. Worked Examples

4.1 Library Search

A library query (a single semantic search over the user’s local documents) executes one chat-model synthesis call and a small number of embedding calls to embed the query. Typical token counts are:

- Chat synthesis: 1500 prompt tokens, 1500 completion tokens
- Embedding: 1 query embedding (1 chunk)

Local compute energy:

$$E_{\text{gen}} = 1500 \cdot 6.7 \times 10^{-5} = 0.100 \text{ Wh} \quad (4)$$

$$E_{\text{prompt}} = 1500 \cdot 3.3 \times 10^{-6} = 0.005 \text{ Wh} \quad (\text{absorbed into rounding below}) \quad (5)$$

$$E_{\text{embed}} = 1 \cdot 1.0 \times 10^{-5} = 1.0 \times 10^{-5} \text{ Wh} \quad (\text{negligible}) \quad (6)$$

$$E_{\text{Mnemne}} \approx 0.10 \text{ Wh} \quad (7)$$

Cloud baseline (single LLM query, PUE-adjusted):

$$E_{\text{cloud}} = 0.34 \cdot 1.4 = 0.476 \text{ Wh} \quad (\approx 0.48 \text{ Wh})$$

Energy saved: $0.476 - 0.10 = 0.376 \text{ Wh}$ ($\approx 0.38 \text{ Wh}$).

Water saved (using rounded local figure of 0.10 Wh; unrounded is 0.105 Wh): $(0.476/1000) \cdot 2.4 - (0.10/1000) \cdot 1.4 = 1.142 - 0.140 = 1.002 \text{ mL}$ ($\approx 1.0 \text{ mL}$).

CO₂e saved: $(0.38/1000) \cdot 380 = 0.14 \text{ g}$.

A typical library query thus saves approximately **0.38 Wh, 1.0 mL of water, and 0.14 g CO₂e** versus a cloud-equivalent.

4.2 Web Search (Agentic)

A web query in Mnemne triggers an eight-call agentic pipeline: query expansion (1 chat call), parallel SearXNG meta-search (no LLM call, local sidecar), parallel page fetching (no LLM, external bytes), per-source distillation (6 chat calls — one per fetched page), and final synthesis (1 chat call). Typical aggregate token counts:

- Total prompt tokens across all calls: $\approx 41,000$
- Total completion tokens: $\approx 6,000$
- Page bytes fetched: $\approx 3 \text{ MB}$

Local compute energy:

$$E_{\text{gen}} = 6000 \cdot 6.7 \times 10^{-5} = 0.40 \text{ Wh} \quad (8)$$

$$E_{\text{prompt}} = 41000 \cdot 3.3 \times 10^{-6} = 0.14 \text{ Wh} \quad (9)$$

$$E_{\text{network}} = 3 \cdot 2.2 \times 10^{-5} = 6.6 \times 10^{-5} \text{ Wh (negligible)} \quad (10)$$

$$E_{\text{Mnemne}} \approx 0.54 \text{ Wh} \quad (11)$$

Cloud baseline (agentic, PUE-adjusted):

$$E_{\text{cloud}} = 2.0 \cdot 1.4 = 2.8 \text{ Wh}$$

Energy saved: $2.8 - 0.54 = 2.26 \text{ Wh}$.

Water saved: $(2.8/1000) \cdot 2.4 - (0.54/1000) \cdot 1.4 = 6.72 - 0.756 = 5.96 \text{ mL}$.

CO₂e saved: $(2.26/1000) \cdot 380 = 0.86 \text{ g}$.

A typical web query thus saves approximately **2.3 Wh, 6.0 mL of water, and 0.86 g CO₂e** versus a cloud-equivalent.

4.3 Aggregate at Scale

For an active user generating 30 queries per day at the typical mix described above (24 library + 6 web), daily savings sum to $(24 \times 0.38) + (6 \times 2.26) \approx 22.7 \text{ Wh}$, $(24 \times 1.00) + (6 \times 5.96) \approx 60 \text{ mL}$, and $(24 \times 0.14) + (6 \times 0.86) \approx 8.5 \text{ g CO}_2\text{e}$. Annualized:

- Energy: $\approx 8.3 \text{ kWh}$
- Water: $\approx 22 \text{ L}$
- CO₂e: $\approx 3.1 \text{ kg}$

For a 100-user organization, annual savings scale linearly to approximately **0.83 MWh, 2,200 L, and 310 kg CO₂e**. These represent estimated displaced consumption under the assumptions documented above, expressible in equivalent units (a year of an LED lamp, a fortnight's worth of daily household showers, a transatlantic flight equivalent) for communicative purposes.

5. Limitations and Confidence Intervals

The methodology presented above is best regarded as a *defensible approximation* rather than a precise measurement. Key limitations follow.

Hardware variability. Apple Silicon GPU power draw varies by chip generation (M1: 8–10 W; M2: 10–14 W; M3: 12–16 W; M4: 14–18 W) and by thermal load. The 12 W coefficient used in Section 3.2 was measured on M2 / M3 systems and has not yet been validated on M4 hardware — we expect it to shift modestly upward (≈ 14 W midpoint) once measured, raising per-token local energy by ≈ 15 –20% on M4 devices. The framework supports a user-supplied override; future revisions will publish per-chip coefficients. Per-token energy estimates currently carry a $\pm 30\%$ range.

Cloud baseline uncertainty. Frontier-model operators have historically not published per-query energy disclosures, with OpenAI’s June 2025 statement of ≈ 0.34 Wh per ChatGPT query [17] the lone recent exception; Anthropic and Google still do not publish per-query figures. The 0.34 Wh single-query midpoint and the 2.0 Wh agentic-query midpoint are derived in Section 3.3 from public open-weight benchmarks and a plausibility argument about agentic call counts. Both carry $\pm 30\%$ to $\pm 50\%$ uncertainty. The relative comparison (Mnemne vs. cloud) is more stable than the absolute Wh figures: at the high end of the cloud range, savings increase; at the low end, they decrease but remain positive for both query types across the full plausible range.

Grid carbon intensity. A single 380 g CO₂e/kWh midpoint loses information for users in grids significantly cleaner (France ≈ 60 g/kWh, Norway ≈ 30 g/kWh) or dirtier (West Virginia ≈ 700 g/kWh, Wyoming ≈ 800 g/kWh) than the US average. Regional lookup is planned but adds runtime complexity and a third-party API dependency.

Local PUE simplifies a complex interaction. As discussed in Section 3.2, local effective PUE varies with climate and season — below 1.0 in heating-dominant cases, above 1.0 in cooling-dominant cases. We default to 1.0 for simplicity. The aggregate effect on annual savings figures is small for users in temperate climates; users in cooling-dominant climates (Phoenix, Houston) are modestly *overcredited*, and users in heating-dominant climates (Boston, Minneapolis) are modestly *undercredited*.

Network attribution is conservative. Section 3.6 attributes only user-side transport energy. A user adopting a full-stack network energy attribution model (e.g., Andrae’s higher coefficient) would shift figures upward proportionally on both local and cloud sides; the *comparison* is comparatively insensitive to that choice because both paths traverse broadly similar public internet and backbone infrastructure, modulo the cloud-internal asymmetries noted in Section 3.6.

Embodied carbon is excluded. Section 3.7 enumerates the cost boundaries; embodied carbon for both user devices and data center hardware is excluded.

Vector-database energy is excluded. Section 3.7 enumerates the LanceDB energy contribution at $\approx 1.5 \times 10^{-4}$ Wh per query, approximately one part in 700 of a typical library query’s total.

Excluded for tractability.

Upstream search-serving asymmetry. Of the costs enumerated in Section 3.7 as excluded from the receipt, the upstream search-serving asymmetry between Mnemne’s metasearch fan-out and a cloud agentic product’s single contracted index is the largest in absolute terms (≈ 1.5 Wh on Hölzle’s 2009 Google figure [20], same order of magnitude as the LLM-inference term itself). It does not appear in Table 1 because it acts on the cancelling-cost side of the comparison rather than on the local or cloud baseline; the reader is referred to the Section 3.7 discussion of mitigating factors and to the receipt-as-disclosure framing of Section 6 for treatment.

Output-quality equivalence is out of scope. The framework measures the environmental cost of *a query*, not the value of its answer. A 3.8B-parameter quantized model running locally and a frontier hosted model are not in general producing equivalent outputs on every task — on hard reasoning, code generation at length, or up-to-the-moment factual recall, the cloud model will often outperform. The accounting we present is valid as a per-query environmental comparison given the user’s choice to use the local tool; it is not a claim that local models are quality- substitutable for cloud frontier models in every case. Quality parity is a separate empirical question, treated elsewhere, and the appropriate framing is “what does the same kind of query cost on each path” rather than “what does a quality-matched query cost.”

The aggregate effect of these limitations is that absolute Wh, mL, and g CO₂e figures should be interpreted with a $\pm 50\%$ uncertainty band. The *relative comparison* — Mnemne saves a meaningful fraction of the cloud equivalent for both query types — holds robustly across the plausible parameter range.

5.1 Sensitivity Analysis

To validate the claim that *savings remain positive across the full plausible parameter range*, Table 1 walks through the worst case (local estimate +30%, cooling-dominant local PUE of 1.33, cloud baseline at the low end of its range, cloud PUE at the low end of the industry range), the midpoint, and the best case (local estimate −30%, cloud baseline at the high end, cloud PUE at the high end) for both query types.

The qualitative claim — Mnemne saves energy versus an equivalent cloud query — holds in the worst case for both query types, with a floor of ≈ 0.09 Wh saved per library query and ≈ 0.61 Wh saved per web query. A reviewer applying their own preferred parameter set can recompute these numbers from the published coefficients in Section 3.

Conditions under which the conclusion would reverse. To make the inequality strict in the other direction — cloud cheaper than Mnemne — would require, *simultaneously*, the cloud per-query baseline falling below ≈ 0.16 Wh (below the lowest published estimate of 0.18 Wh, the bottom of Oviedo et al.’s IQR [18]) *and* the local on-device PUE saturating at the cooling-dominant ceiling of 1.33 for every query in the user’s annual mix. Both reversal thresholds are computed at $\text{PUE}_{\text{cloud}} = 1.10$ (hyperscaler best-in-class), consistent with the worst-case assumptions in Table

Scenario	Local (Wh)	Cloud (Wh)	Saved (Wh)
Library, worst case	0.173	0.262	0.089
Library, midpoint	0.100	0.476	0.376
Library, best case	0.070	0.698	0.628
Web, worst case	0.934	1.540	0.606
Web, midpoint	0.540	2.800	2.260
Web, best case	0.378	4.108	3.730

Table 1: Sensitivity analysis. Savings remain positive in all six scenarios. The worst case stacks three pessimistic assumptions simultaneously: local energy 30% above midpoint, local PUE at the cooling-dominant ceiling of 1.33, and cloud energy 30% below midpoint paired with the low end of the cloud PUE range (1.10, hyperscaler best-in-class). The best case stacks the opposite three: temperate-climate $\text{PUE}_{\text{local}} = 1.0$, local estimate 30% below midpoint, cloud energy 30% above midpoint, and cloud PUE at the high end of the industry range (1.58, average across all surveyed data centers). In heating-dominant climates ($\text{PUE}_{\text{local}} < 1.0$ from waste heat displacing space heating) savings would be higher still.

1. We are not aware of a credible parameter set that satisfies both conditions, so we treat the qualitative claim as falsifiable but not falsified. For the agentic-pipeline comparison the margin is wider still: the cloud agentic baseline would have to fall below ≈ 0.85 Wh (less than three single-shot calls, against the 5–8-call plausibility argument of Section 3.3.2) for Mnemne to lose. A genuine falsification would most plausibly come from a future disclosure that frontier providers serve at roughly the local-Mac batch-1 figure, which would require an architectural shift we have no evidence is occurring.

6. Discussion: Sustainability-as-Receipts

The methodology described above is engineered for a specific user experience: rendering each AI query as an itemized environmental receipt analogous to a financial transaction record. We argue this approach has three properties absent from aggregate sustainability disclosures.

Specificity permits direct methodological inspection. A statement of the form “this query saved 2.3 Wh” is associated with a defined computation over coefficients listed in this paper, and a technically sophisticated user can recompute it. Aggregate corporate sustainability statements typically do not expose computation at that level of granularity, regardless of their underlying accuracy.

Per-action accumulation aggregates user-visible quantities. A running total (“you have saved 8,300 Wh this year, equivalent to a smartphone recharged ≈ 600 times”) presents the cumulative estimated savings in concrete units. This is structurally analogous to step counters, fuel-economy displays, and bank-balance tickers, which present transactional aggregates of approximate measurements to support user decision-making.

Methodology transparency permits independent recomputation. Publishing the coeffi-

cients, ranges, and computational form (this paper) lets independent reviewers recompute any line of the receipt. The methodology is explicitly falsifiable and designed for independent recomputation; we expect to revise it as more accurate measurements become available.

We label the resulting pattern *sustainability-as-receipts*: a disclosure approach that treats environmental impact reporting as a transactional artifact accompanying user actions. Similar patterns exist in adjacent domains — food provenance (“farm to fork”), data privacy (“network log”), financial transactions (“itemized expense”) — united by the working hypothesis that transaction-level disclosure provides information that aggregate disclosure does not.

6.1 On the Sufficiency of Approximate Receipts

A natural objection: per-query environmental estimates carry uncertainty bands of $\pm 30\text{--}50\%$ (Section 5). Does transaction-level disclosure of approximate quantities provide meaningful information?

We note three established cases in which approximate transaction-level disclosure is treated as informative despite explicit measurement uncertainty:

- *Nutrition labels.* FDA-mandated calorie counts on packaged food carry tolerances of $\pm 20\%$, with even wider practical variance for restaurant menu disclosures. Public-health research treats these labels as behaviorally meaningful notwithstanding the tolerance.
- *Vehicle fuel economy.* EPA mpg estimates are well known to diverge from individual-vehicle observed economy; consumers and regulators nonetheless treat them as informative for cross-vehicle comparison and aggregate fleet planning.
- *Smartphone battery indicators.* Per-app energy disclosures in iOS and Android settings panels carry significant attribution uncertainty; users nonetheless act on them when deciding which apps to remove or background-restrict.

In each case, the per-transaction signal is treated as informative relative to no signal at all, and the uncertainty is bounded and disclosed. We argue per-query AI environmental receipts sit in the same category: they are not precision instruments, but the alternative — aggregate annual reporting that the user cannot attribute to their own actions — provides strictly less actionable information.

7. Generalization

While the reference implementation here is specific to Mnemne, the framework parameterizes naturally. A different local-first AI application would need:

- A measured or estimated per-token energy coefficient for its chosen model on its target hardware (Section 3.2).
- A per-query-type cloud baseline derived from the closest cloud-product analog (Section 3.3).

- The water and carbon coefficients of Section 3.4–3.5 (which are properties of the grid and data center operators, not the application).

The framework’s applicability is not limited to Apple Silicon. An Ollama deployment on Linux with an NVIDIA GPU, for example, would substitute a different per-token coefficient (measured via `nvidia-smi` rather than `powermetrics`; the measurement methodology is materially different in that `nvidia-smi` reports board power directly while `powermetrics` reports SoC-level power that must be attributed to the GPU partition). The receipt formula and the cloud-baseline derivation in Section 3.3 are unchanged. The methodology of this paper is therefore reusable. We expect future work to refine the per-token coefficients as more accurate measurements become available across hardware generations, and to extend the cloud-baseline derivations as frontier-model operators disclose more inference-time data.

8. Conclusion

We have documented the per-query environmental accounting methodology used in Mnemne for local-first AI queries: coefficient derivations, explicit cloud-baseline derivation, climate-aware PUE treatment, network-energy framing, and explicit cost-boundary enumeration. Worked examples for two query types (library, agentic web) and a limitations section with $\pm 50\%$ uncertainty bands accompany the framework. We argued that per-query receipts are an underappreciated trust mechanism for AI sustainability, structurally analogous to itemized financial receipts and to network-log-based privacy disclosures.

The full implementation, including the runtime that produces these receipts and exposes them to users, ships in Mnemne 0.9.11 and later releases.

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